



Cross Architecture Approach to AI

John Cuneo - Worldwide AI Solutions Engineer
AIHUB-1009



Webex App

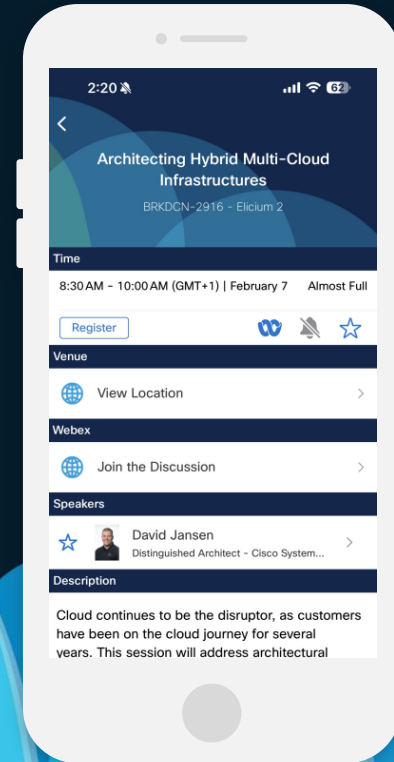
Questions?

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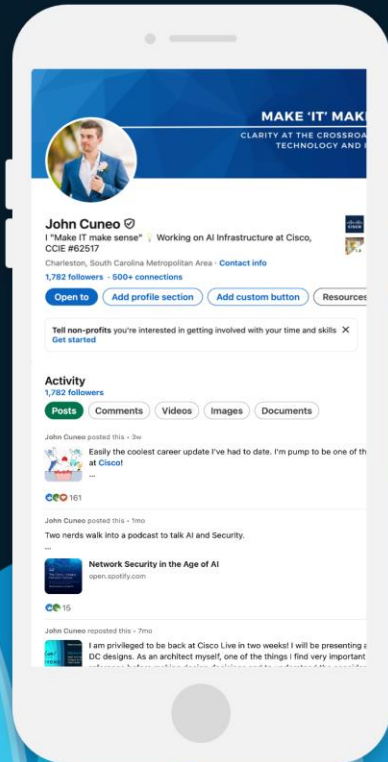
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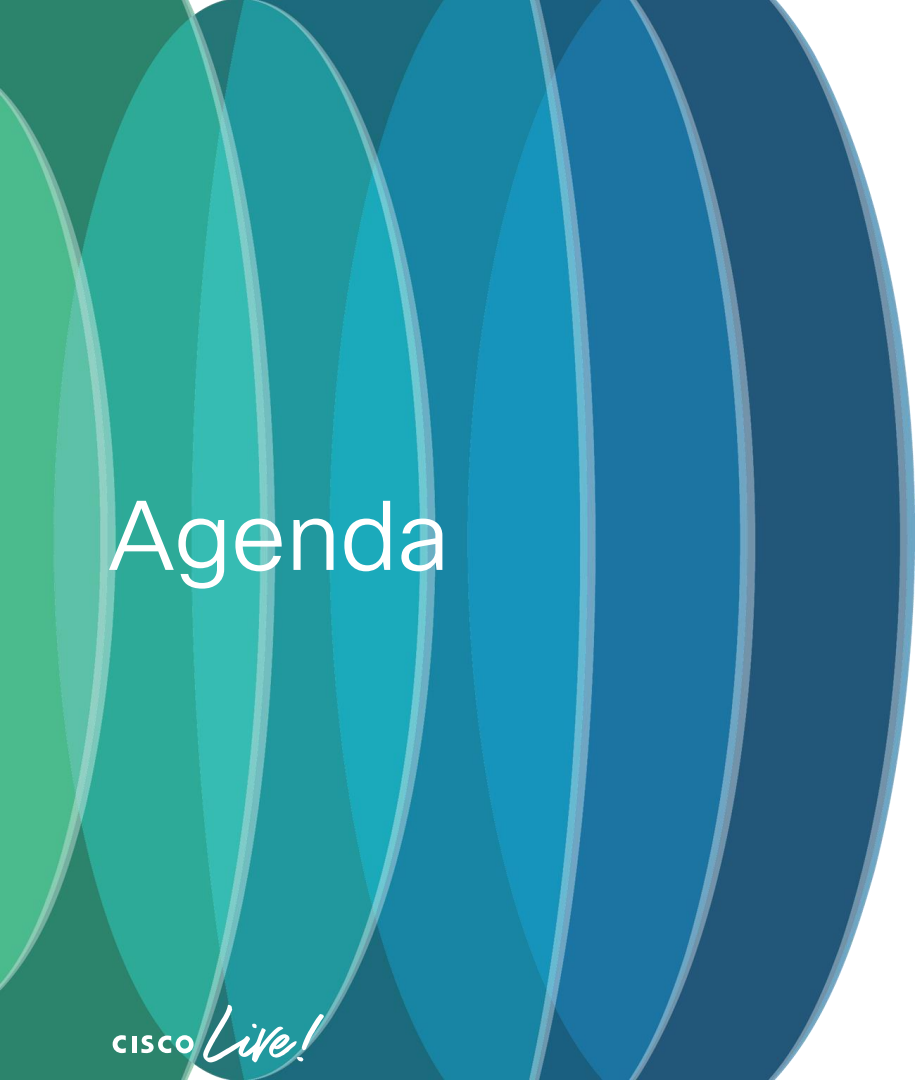
Who's John

What I do...

- Global technical AI go-to-market
- Strategic customer engagement
- Executive briefing
- Eat, breathe and dream GenAI for 2 years
- Lifelong fantasy nerd & gamer



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Agenda

- Looking Back: the Journey to the current AI Era
- Cross-Architecture Considerations
- AI Readiness
- Cisco For AI

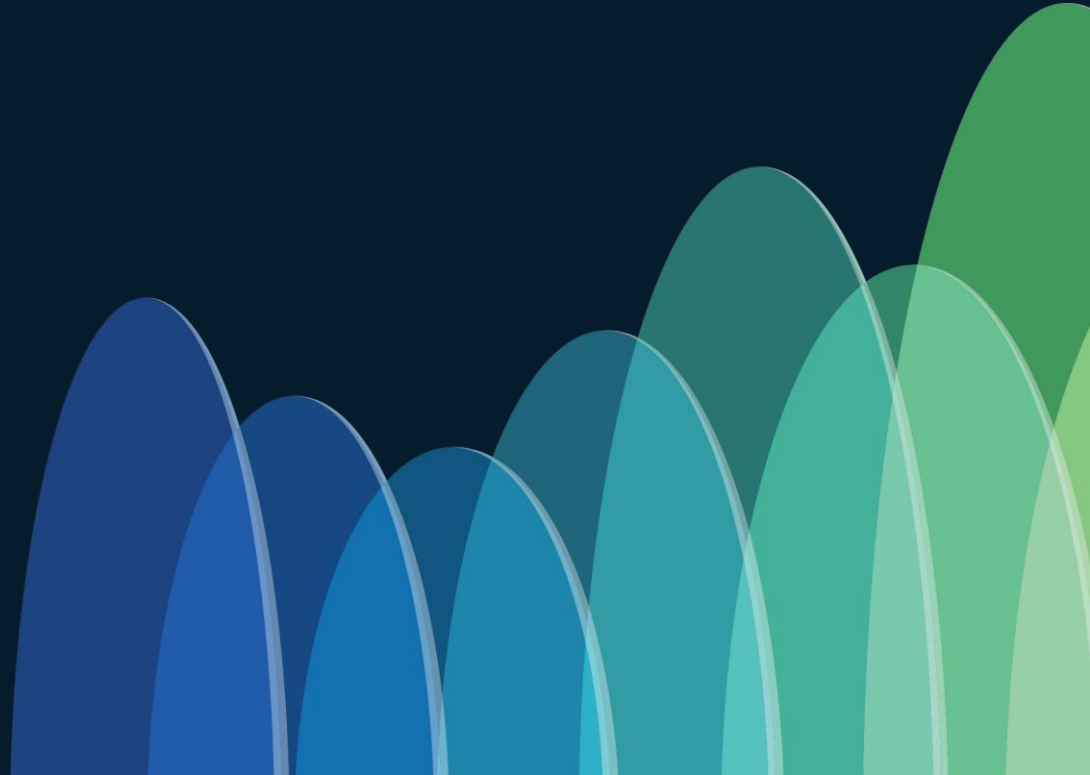
Learning Objectives

By the end of this session, you'll be able to:

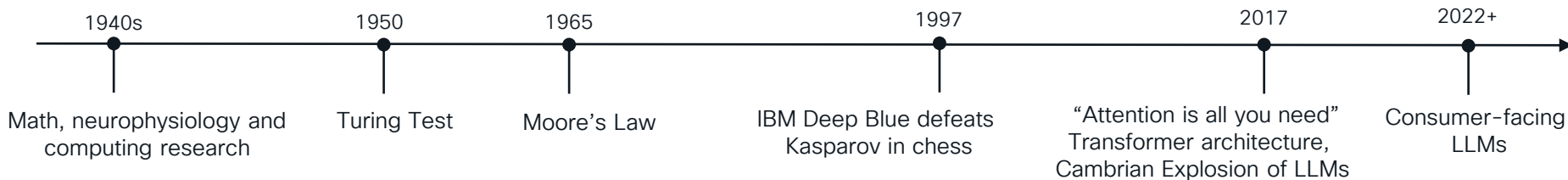
- Approach your AI strategy with key considerations in mind
- Describe cross-architecture AI Readiness
- Understand Cisco's strategy for AI



The AI Era



The AI Era – How did we get “here”?



AI

Machine Learning

IBM 701 computer plays checkers

Computational models

Growth in availability of data

Probability models

Statistical analysis techniques

Deep Learning

Artificial Neural Networks

Natural Language Processing

Image recognition

Google Translate

Gen AI

BloombergGPT

ChatGPT

Foundational Models

Domain-specific Models

Claude

Code Generation

Image Generation

Keeping Pace

AI capabilities continue to develop...

Basic Text Generation

- Basic tasks with “emergent” results
- Writing & Coding
- Text Summarization
- Translation

Augmented Generation

- Integrated store of information to enhanced prompt
- Then execute basic text generation

Tool Calling

- Ability to execute custom functions/API calls
- Query for structured source of truth
- Interact with existing applications

Advanced Reasoning

- Simulate human thought process via recursive prompting “chain of thought”
- Deconstruct request
- Plan the approach
- Execute based on the plan
- Validate if response makes sense

Token Utilization

GPT Launch

Unlocking new use-cases

What *really* changed?

General availability of consumer and enterprise grade models



010110
110010
001011



Scientific research realm

Consumer and business realm

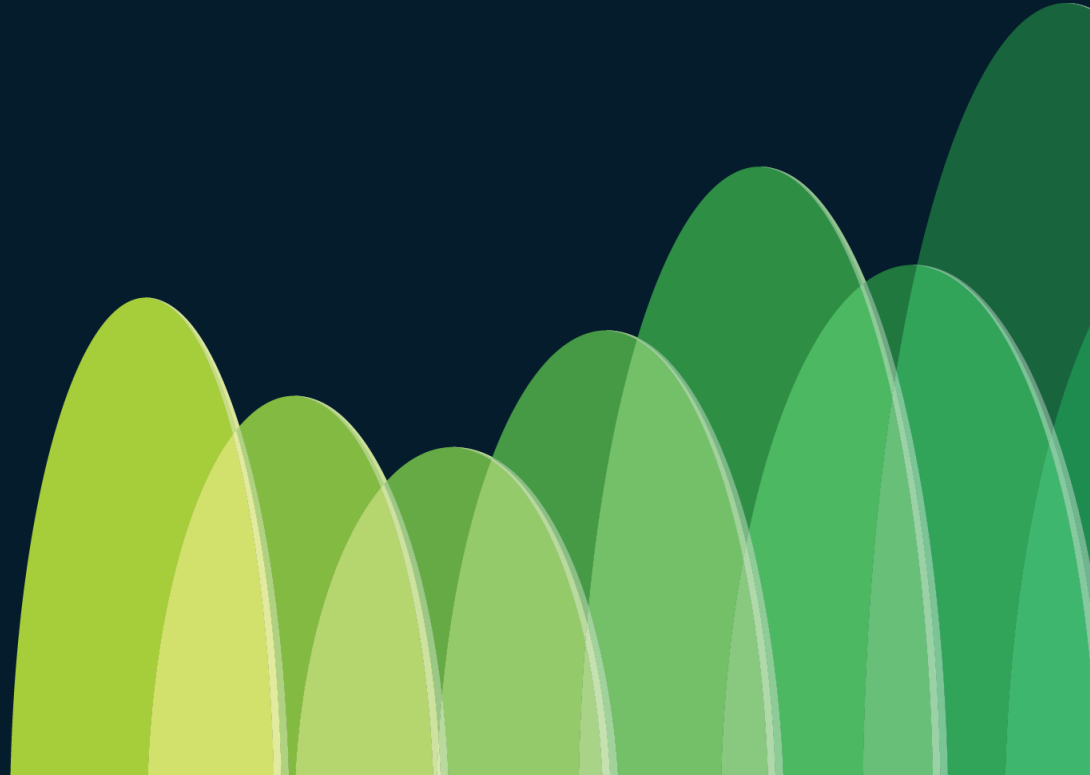
Challenges for Organizations

Data Governance

Scalable Systems

Risk Management

Cross Architecture Considerations



Differentiate Features from AI vs Systems for AI

Features from AI

Background noise removal
Self-service tools
Predictive analytics in t-shoot
AI assistants

Typically delivered via SaaS

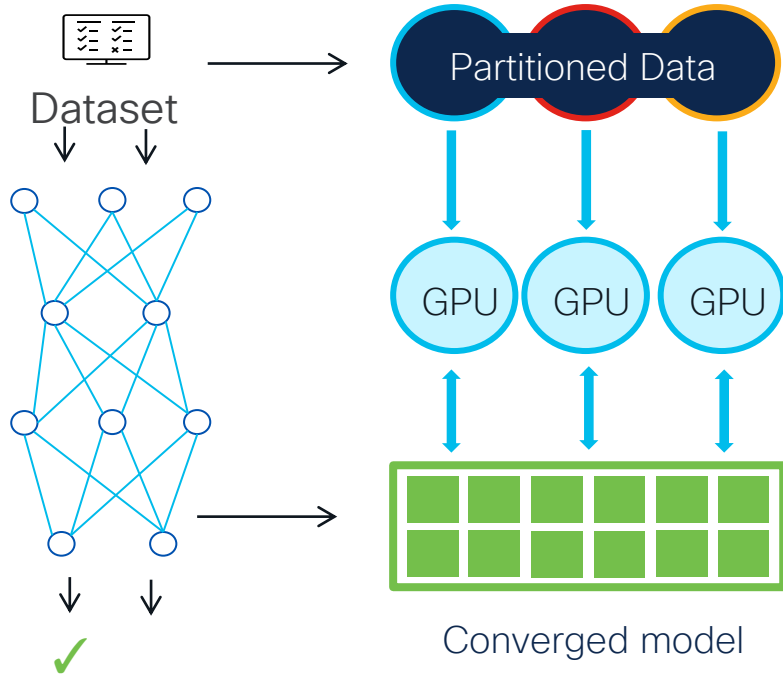
Systems for AI

Deep Neural Network model design
Back-end and front-end networks
Data storage
Deep Learning frameworks and SDKs

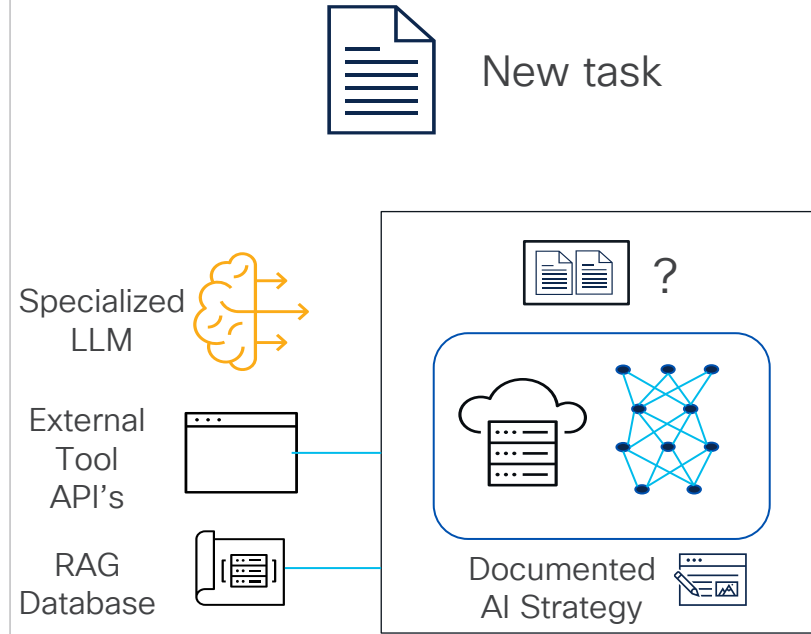
On-prem, Cloud

What does a system for AI entail?

Training & Fine Tuning

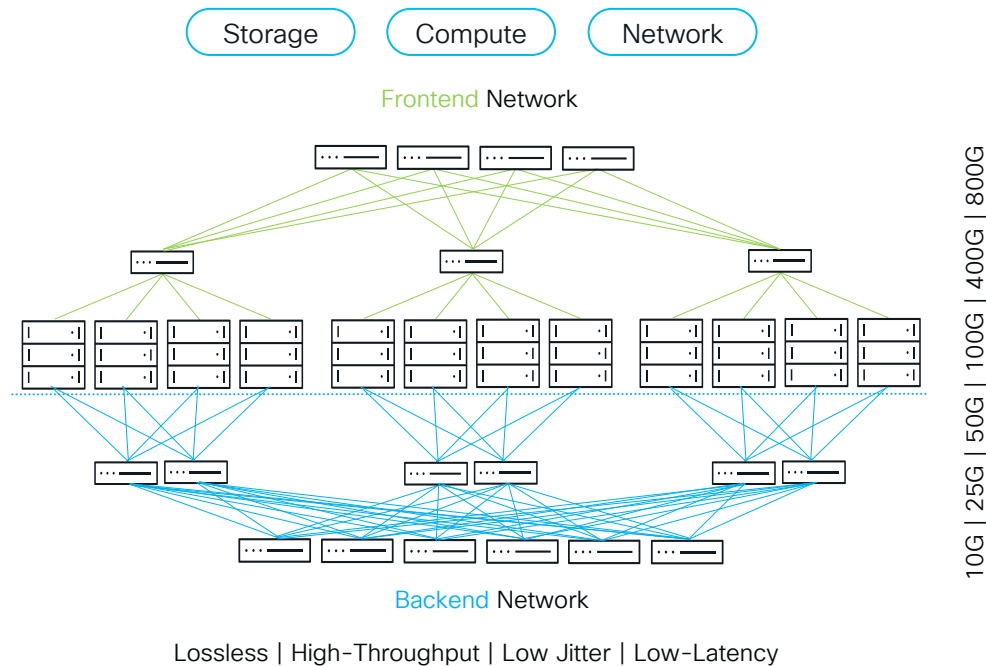


Inferencing



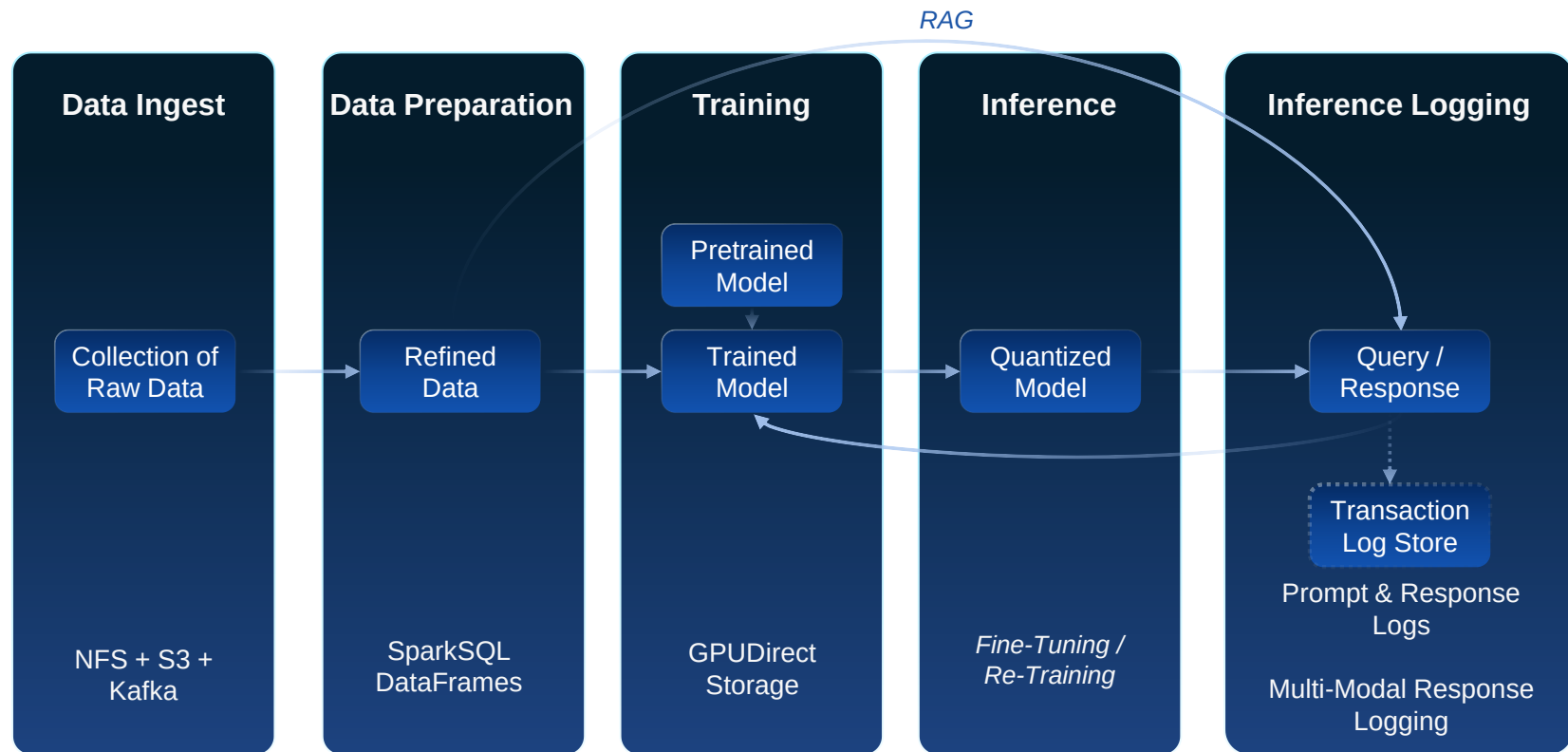
What does infrastructure for AI entail?

- High throughput, lossless fabric for training
- Can require extensive power and cooling
- Training and Inference can differ in compute density and utilization



It's all about the **data**

AI Data Pipeline



Moving the data

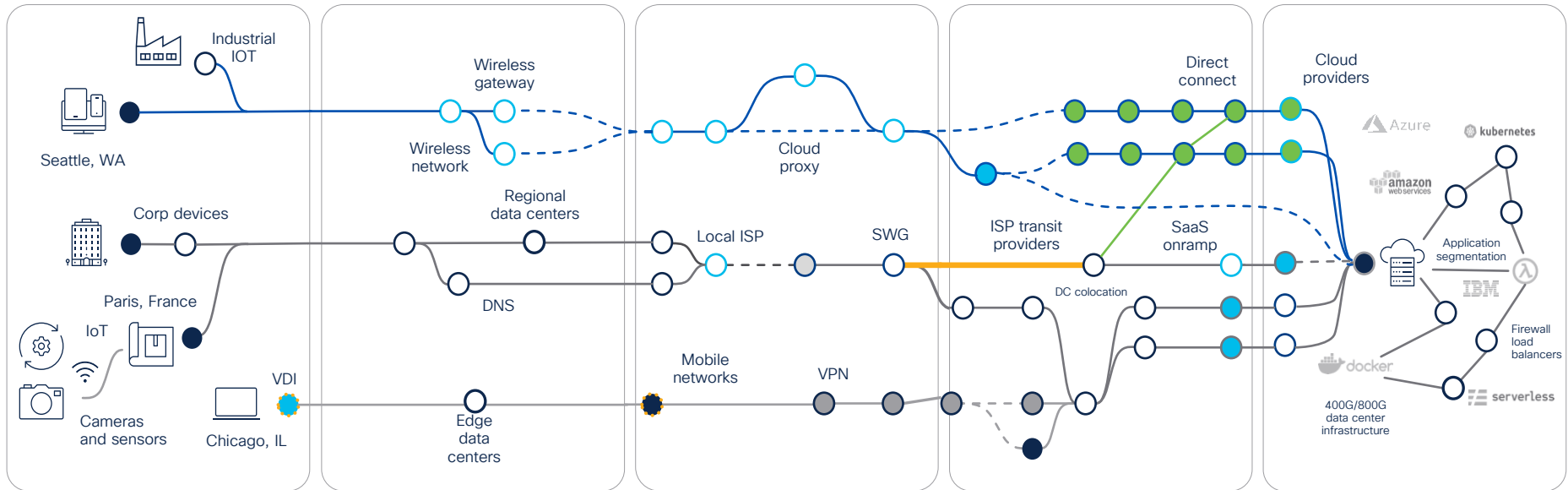
People, places and things

Access networks

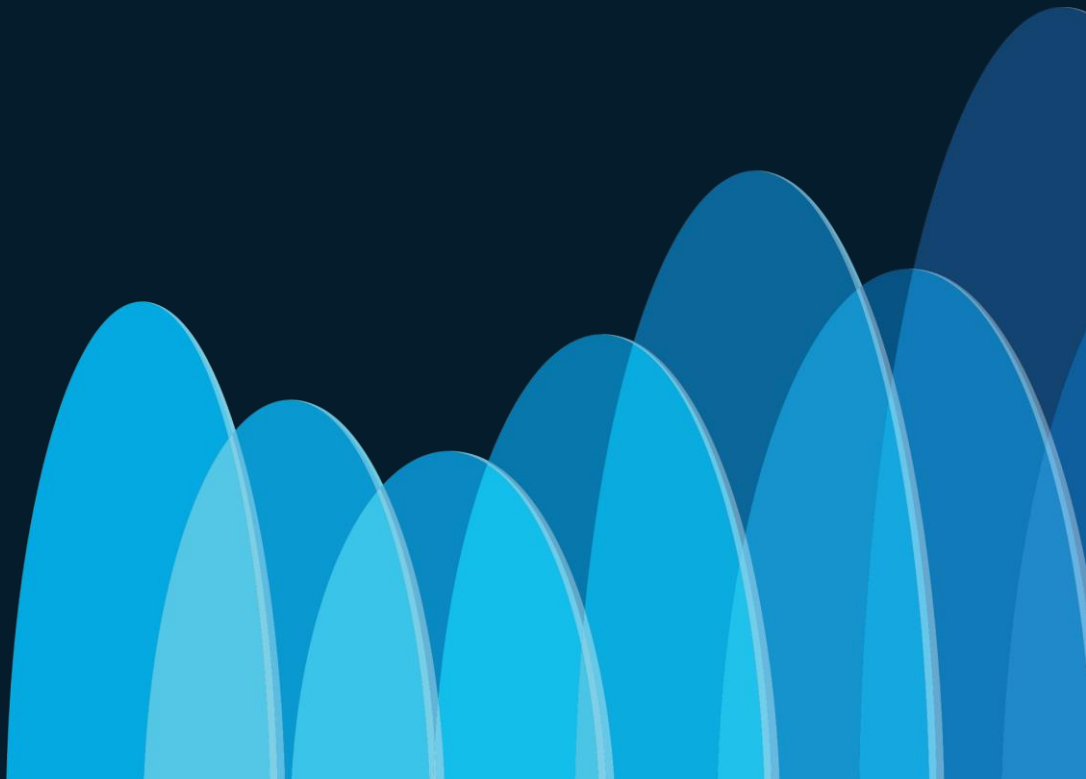
Network services

Cloud connectivity infrastructure

Apps (data center, cloud)

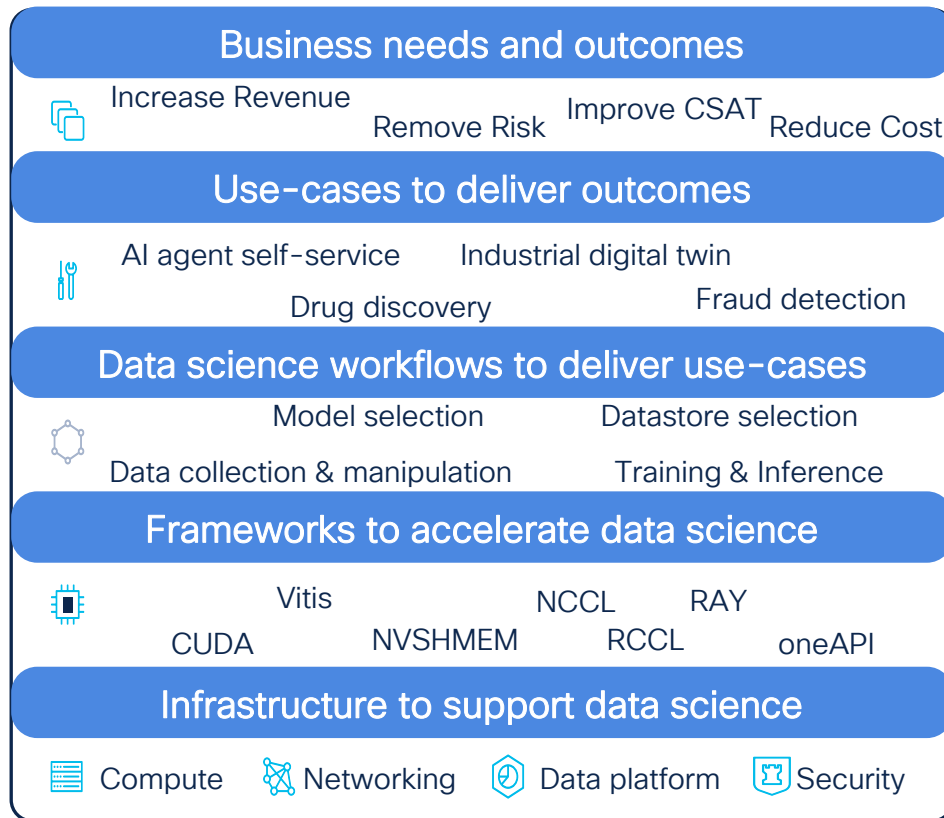


How can I start?



Initial Questions to Define Cross Architecture Requirements

- What is the outcome?
- What is the use case?
- What data do I need?
- What model will deliver that use case?
- What is my model size?
- How will I manage resource utilization?
- What security policies will I need to implement?



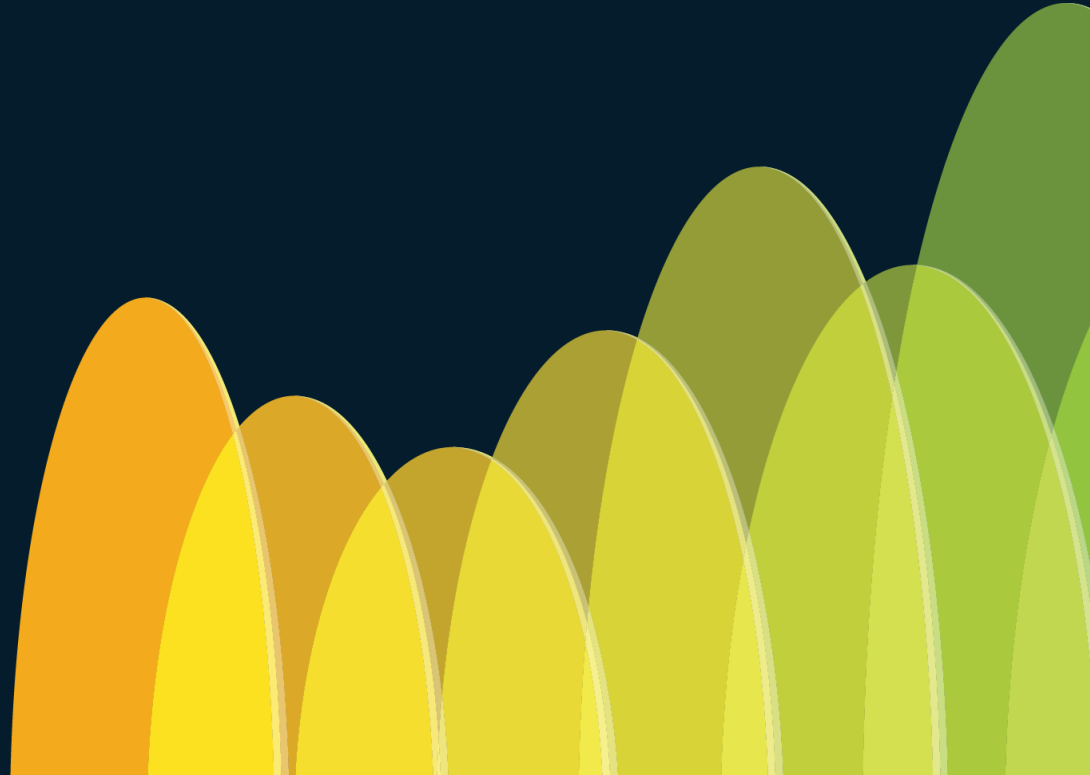
Considerations for AI-Readiness

From the Cisco [“Journey to AI”](#) guide



- Form an AI committee and establish clear Governance
- Project the required infrastructure
- Ensure hygiene, quality, and reliability of in-house and external data sources
- Implement mechanisms to detect bias and checks on AI algorithms
- Establish proficiency for responsible usage of AI

Cisco for AI



Cisco's AI approach

Using AI to drive operational efficiency for **better experiences**



AI-Native IT operations

Networking | Assurance | Collaboration
Security | Responsible AI

Enabling **Your infrastructure** to support adoption of AI applications

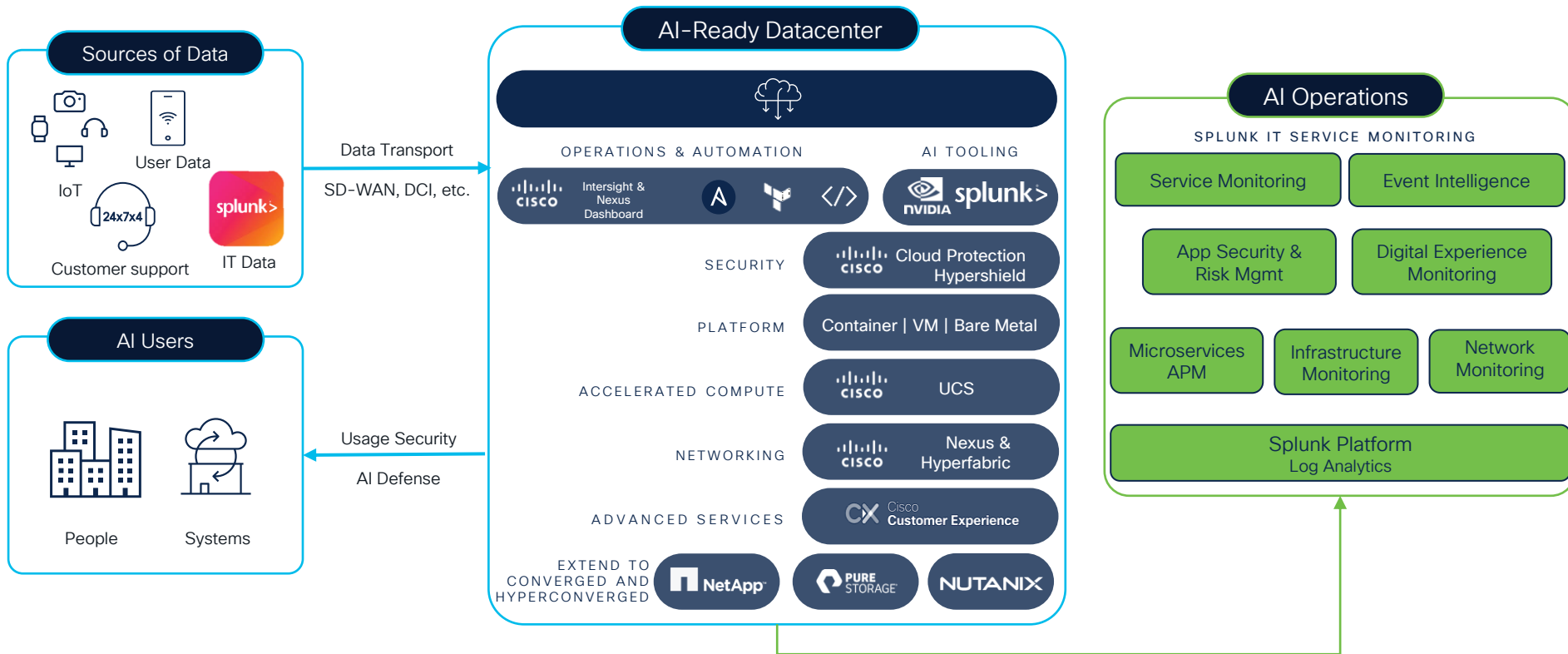


AI-Ready infrastructure

Data Center Networking | Compute |
Storage | Security | Trusted AI

Cisco End-to-End AI

Following the Data

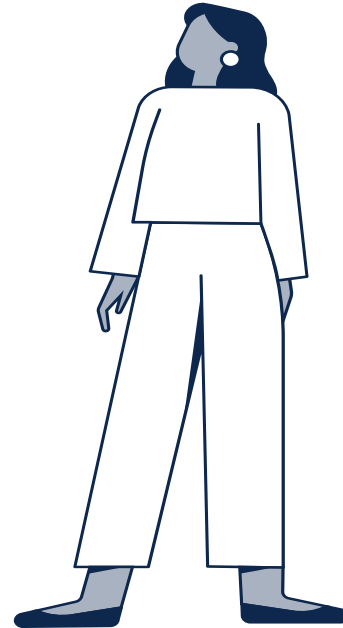


What should I do
next?



Next Steps

- Determine your AI readiness across your environment
- Continue building your AI literacy
- Develop a cross-architecture roadmap for AI
- Start small and have fun with it



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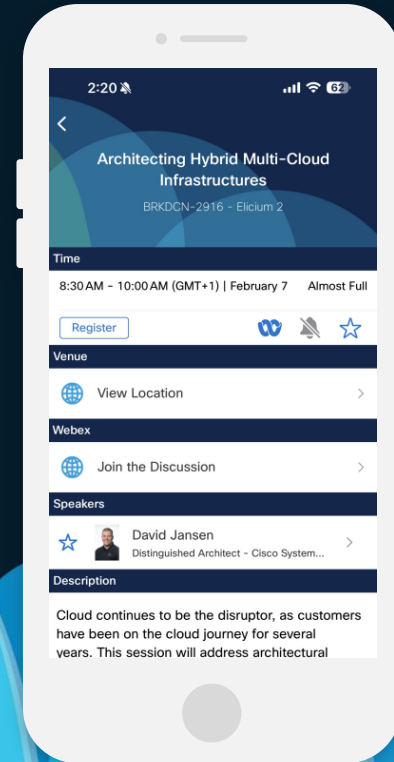
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Connect on LinkedIn: <https://www.linkedin.com/in/cuneojohn/>
Contact me at: johcuneo@cisco.com

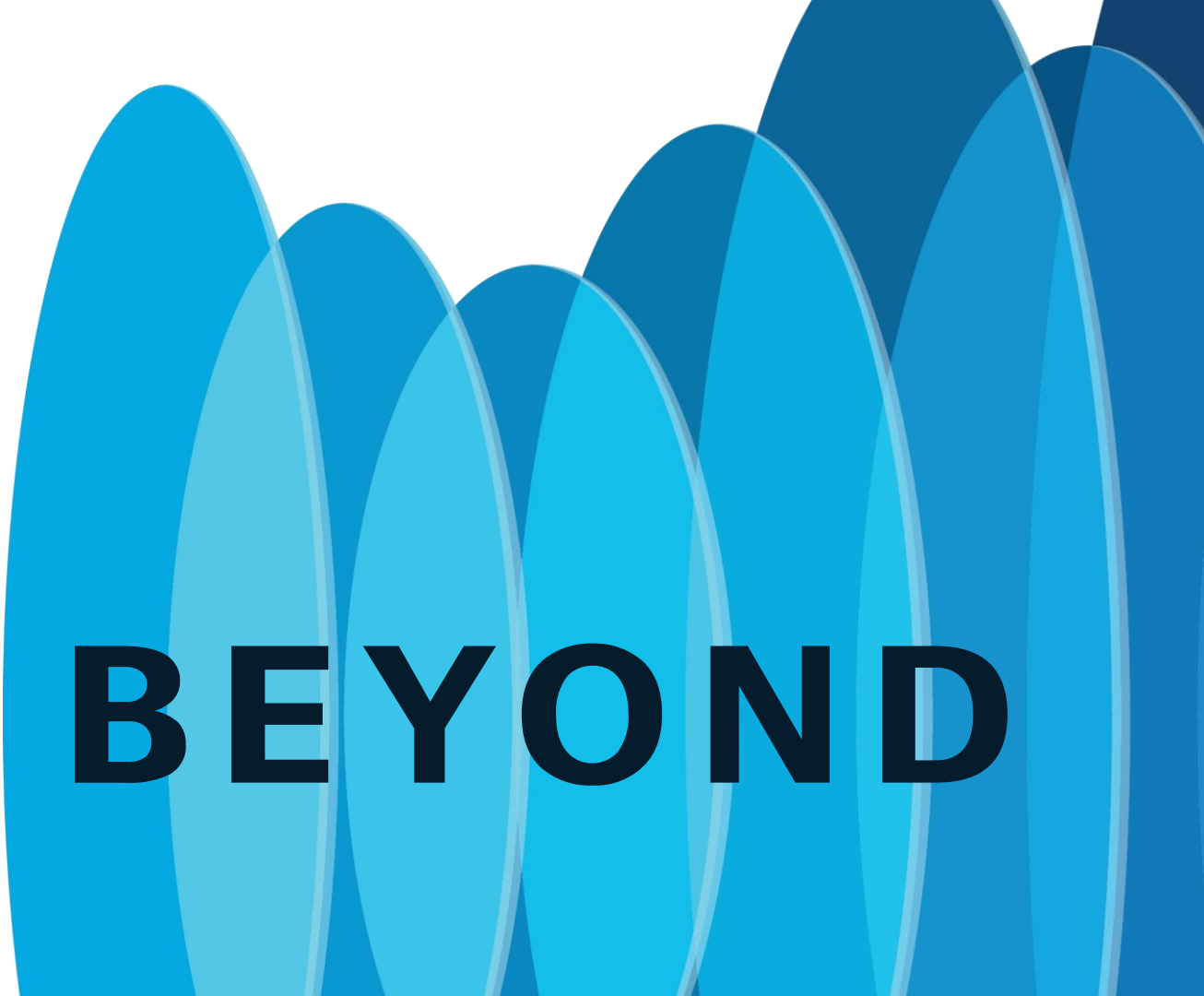


Thank you



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GO BEYOND

The background of the slide features a series of overlapping, teardrop-shaped elements in various shades of blue, ranging from light sky blue to deep navy blue. These shapes are arranged in a way that creates a sense of depth and movement, resembling a stylized horizon or a series of waves. The overall composition is clean and modern, with a focus on the bold text 'GO BEYOND'.